

Multiple Choice

1. **Answer:** B

Options: A) Rocco read six books in two months.; B) Rocco hiked six miles each day for two days.; C) Rocco spent six hours on homework for two days.; D) Rocco has six siblings and two of them are girls.; E) Rocco had a total of six tennis balls in two cans.

Note: Correct option: B - Rocco hiked six miles each day for two days.

2. **Answer:** E

Options: A) 2.5; B) 1.7; C) 1.9; D) 2.2; E) 1.8

Note: Correct option: E - 1.8

3. **Answer:** A

Options: A) 1; B) 8; C) 0; D) 7; E) 5

Note: Correct option: A - 1

4. **Answer:** A

Options: A) 25; B) 35; C) 17; D) 13; E) 9

Note: Correct option: A - 25

5. **Answer:** A

Options: A) 110 parts; B) 245 parts; C) 770 parts; D) 220 parts; E) 140 parts

Note: Correct option: A - 110 parts

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Matches the MMLU-Pro answer key.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '100 and 299'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.8'.

10. **Answer:** False
Options: A) True; B) False
Note: Actual answer is '-1.047'.

Long Form Response

11. **Answer:** The tens digit cannot be 2, 5, or 8 because otherwise, the sum of the digits, and thus the number itself, is divisible by 3. The remaining possibilities are 17, 37, 47, 67, 77, and 97. Only $77 = 7 \cdot 11$ is composite, so there are 5 two-digit prime numbers that have a units digit of 7.
Note: Students should show all steps in their mathematical solution.
12. **Answer:** There are 5 steps to the right, and 2 steps up. These 7 steps can be made in any order, so the answer is $72 = 7 \times 62 \times 1 = 21$.
Note: Students should show all steps in their mathematical solution.